

Multilingual Clinical Symptom Understanding and Department Routing Model with Conversational Integration

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Abstract

India’s linguistic diversity presents a major challenge in building accessible digital healthcare systems. Most existing AI-driven medical assistants are limited to English and lack domain-specific clinical understanding.

This project develops a multilingual medical symptom routing system that retrieves the most relevant disease and clinical department from a curated dataset based on symptom similarity. The system is trained on a curated dataset of approximately 169,000 symptom-description sentences derived from a disease-symptom medical dataset.

To demonstrate real-world applicability, the system integrates a custom medical inference pipeline with a conversational chatbot interface powered by Google Gemini. The chatbot manages natural dialogue and user interaction, while the backend model performs symptom-based department routing using similarity-based retrieval.

1 Introduction

Recent advances in artificial intelligence have accelerated the adoption of digital tools for healthcare assistance, particularly in preliminary patient guidance and symptom-based department routing. However, most existing AI-based healthcare systems remain English-centric and rely heavily on general-purpose language models with limited domain-specific medical knowledge.

In linguistically diverse regions such as India, patients frequently describe symptoms using regional languages or mixed-language expressions (such as Hinglish). This makes accurate interpretation of patient complaints challenging for conventional models trained primarily on formal medical text.

Hospitals also face increasing patient loads, where efficient routing of patients to the correct medical department plays an important role in improving healthcare delivery. Automated systems

that can understand symptom descriptions and map them to relevant specialties can significantly improve early-stage patient guidance.

To address these challenges, this work introduces a hybrid framework integrating a custom medical NLP pipeline with a conversational interface powered by Google Gemini.

2 Literature Survey

Prior work in medical natural language processing has focused on clinical entity recognition, diagnosis prediction, and automated triage using transformer-based architectures. Biomedical adaptations of BERT, including BioBERT and ClinicalBERT, have demonstrated strong performance on structured clinical notes.

However, these models are primarily trained on hospital-generated data and often fail to generalize to patient-authored symptom descriptions, which tend to be informal and linguistically diverse.

Recent research in Indian-language healthcare NLP highlights challenges related to code-mixing, regional expressions, and limited annotated datasets. Multilingual transformer models such as XLM-R, MuRIL, and IndicBERT enable cross-lingual representation learning but require domain-specific datasets for effective clinical applications.

Conversational healthcare assistants increasingly rely on large language models for dialogue management. However, studies indicate that LLM-based systems may produce inconsistent medical outputs when used directly for triage tasks. Hybrid systems separating conversational interaction from structured medical reasoning have therefore become an emerging research direction.

Motivated by these limitations, this work proposes a hybrid architecture combining conversational interaction through Gemini with a structured symptom-to-department routing model.

3 Problem Statement

Patients often describe symptoms using informal language, regional dialects, or mixed-language text. Mapping these symptom descriptions to the appropriate medical department is a critical step in preliminary healthcare triage.

Existing systems face several limitations:

- Lack of datasets mapping symptoms directly to clinical departments
- Limited support for multilingual patient inputs
- Over-reliance on large language models for medical reasoning

This project addresses these limitations by constructing a curated symptom-to-department dataset and developing a multilingual routing model capable of interpreting natural language symptom descriptions.

4 Objectives

The primary objectives of this project are:

1. Build a model capable of mapping symptom descriptions to clinical departments
2. Support multilingual patient symptom inputs
3. Construct a curated dataset linking diseases, symptoms, and departments
4. Integrate the model into a conversational chatbot interface

5 System Architecture

This section was newly added.

The proposed system follows a hybrid architecture combining a conversational interface with a structured medical inference pipeline.

- **Language Processing Layer:** User inputs are first processed through language detection and translation components to normalize multilingual inputs.
- **Symptom Matching Engine:** Natural language symptom descriptions are converted into vector representations and compared with entries in the curated dataset.
- **Department Routing Module:** The system identifies the most similar symptom descriptions and retrieves the associated disease and medical department.

- **Conversational Interface:** Google Gemini manages the conversational flow, collects symptom descriptions, and presents model outputs in a user-friendly format.

- **Integration Layer:** REST APIs connect the chatbot with the backend medical inference system.

6 Methodology

This section was newly added.

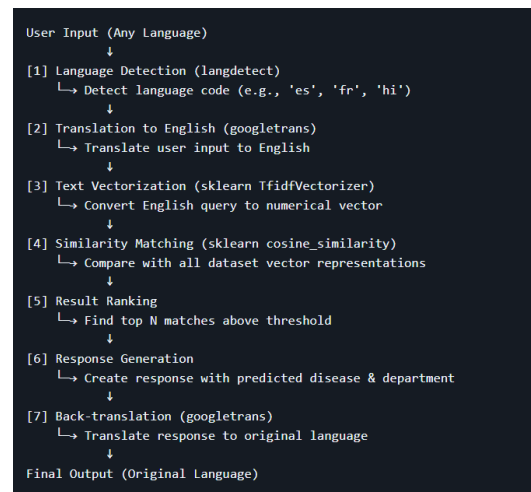


Figure 1: Pipeline of the multilingual symptom routing system. The system detects the input language, translates the query to English, converts it into TF-IDF vectors, retrieves similar symptom descriptions using cosine similarity, and generates a response which is translated back to the user's language.

6.1 Dataset Construction

The training dataset is derived from a publicly available **Disease-Symptom dataset** containing:

- 773 unique diseases
- 377 symptoms
- 246,945 records

Each record associates a disease with a set of symptoms encoded as binary features.

6.2 Sentence Generation

To enable natural language processing, symptom vectors were converted into descriptive patient-style sentences. For example:

- “I have fever and persistent cough.”
- “I am experiencing headache and dizziness.”

After preprocessing and filtering, **169,888 symptom-description sentences** were generated.

6.3 Department Mapping

Each disease was mapped to the appropriate clinical department. Initial suggestions were generated using GitHub Copilot and subsequently verified to ensure clinical relevance.

Examples include:

- Pneumonia → Pulmonology
- Migraine → Neurology
- Psoriasis → Dermatology

The resulting dataset contains three fields:

- Disease
- Symptom Description Sentence
- Medical Department

6.4 Dataset Statistics

The original Disease–Symptom dataset contains **246,945 records** describing associations between diseases and symptoms. Each record is represented using **377 binary symptom features**, indicating the presence or absence of specific symptoms for a given disease. The dataset covers **749 unique diseases**.

After preprocessing, sentence generation, and deduplication, the final curated dataset contains **169,888 unique samples**. Each entry consists of a natural language symptom description paired with its corresponding disease and medical department.

6.4.1 Department Distribution

The generated dataset contains symptom descriptions mapped to **21 medical departments**. The distribution of the most frequent departments is shown below:

- General Medicine: 50.01% (84,966 records)
- Gastroenterology: 7.03% (11,935 records)
- Orthopedics: 4.74% (8,055 records)
- Psychiatry: 4.57% (7,759 records)
- Other 17 departments: 25.65%

The high proportion of samples in General Medicine reflects the broad symptom coverage associated with general clinical consultations.

6.5 Baseline Model

The baseline system uses a similarity-based retrieval approach for symptom matching.

First, user input language is detected using the *langdetect* library. If the input is not in English, it is translated into English using the *googletrans* translation library.

The normalized English input is converted into numerical vectors using **TF–IDF vectorization**. TF–IDF assigns weights to words based on their importance in the document and across the dataset. TF–IDF was chosen for the baseline system due to its computational efficiency, interpretability, and effectiveness in identifying symptom keywords within short clinical descriptions.

Cosine similarity is then computed between the query vector and all dataset sentence vectors. The highest similarity scores correspond to the most relevant symptom descriptions.

The disease and department associated with the top matching sentences are returned as the model prediction.

6.6 Implementation Libraries

The baseline implementation uses the following libraries:

- **langdetect** for language identification
- **googletrans** for multilingual translation
- **scikit-learn** for TF–IDF vectorization and cosine similarity
- **pandas** for dataset handling
- **numpy** for numerical operations

7 Evaluation

This section was newly added.

To evaluate the baseline system, symptom queries were tested against the curated dataset to measure department prediction accuracy.

7.1 Evaluation Setup

The evaluation considered:

- Correct disease retrieval
- Correct department prediction
- Multilingual input handling

Accuracy was measured by comparing the predicted medical department returned by the similarity matching system with the ground truth department label associated with the matched dataset entry.

Cosine similarity thresholds were used to filter low-confidence matches.

7.2 Baseline Performance

Metric	Result
Department Prediction Accuracy	75–85%
Average Query Latency	0.5–2 seconds
Supported Languages	50+

Table 1: Baseline model performance

```

ananyasingla@ananyas-MacBook-Pro:~/Advanced_NLP_Project % python3 medical_chatbot.py
Loading dataset...
Building TF-IDF vectors (this may take a moment)...
✓ Dataset loaded: 169888 records
✓ TF-IDF matrix built: (169888, 5000)
✓ Chatbot initialized and ready!

=====
MULTILINGUAL MEDICAL SYMPTOM CHATBOT
=====
Welcome! I can understand and respond in multiple languages.
Describe your symptoms in any language you prefer.
Type 'quit' or 'exit' to end the conversation.

. You: I am experiencing emotional symptoms, nausea, and uterine contractions.
(1/5) Detecting language...
✓ Detected language: English
(2/5) Translating to English...
✓ English translation: "I am experiencing emotional symptoms, nausea, and uterine contractions."
(3/5) Searching for matching diseases...
✓ Found 3 matches
(4/5) Translating response to English...
✓ Response translated
(5/5) Complete

=====
■ Top Match: placental abruption
■ Department: Gynecology
■ Confidence: 98.6%
● Response:
Based on your symptoms, the most likely conditions are:
1. placental abruption (Gynecology) - Confidence: 98.6%
2. hypothermia (General Medicine) - Confidence: 68.6%
3. testicular torsion (Urology) - Confidence: 56.8%
The top recommendation is: placental abruption (Department: Gynecology)

. You:

```

Figure 2: Example execution of the multilingual medical symptom chatbot. The system detects the input language, retrieves similar diseases using TF-IDF similarity, and returns the predicted disease and clinical department along with confidence scores.

The similarity-based retrieval model performs reliably for symptom descriptions that closely resemble dataset entries. However, performance decreases for highly paraphrased or ambiguous queries.

8 Chatbot Integration

The trained medical inference pipeline is deployed as a REST API.

The chatbot powered by Google Gemini handles:

- conversational interaction
- user symptom collection
- response presentation

When a user describes symptoms, the chatbot sends the query to the backend system. The backend computes similarity scores and returns the most relevant disease and department. Gemini then presents the output in conversational form.

9 Example Interaction

An example interaction with the system is shown below:

- **User Input:** "I have fever, cough, and chest pain."
- **Predicted Disease:** Pneumonia
- **Predicted Department:** Pulmonology

The chatbot presents this information conversationally while advising the user to consult the appropriate medical department.

10 Novelty and Innovation

This project introduces several key contributions:

- Construction of a large symptom-to-department dataset
- Conversion of structured symptom vectors into natural language sentences
- Multilingual symptom understanding using translation-based normalization
- Hybrid architecture combining conversational LLMs with structured medical reasoning

11 Expected Outcomes

The project delivers:

- A curated dataset of symptom descriptions linked to diseases and departments
- A similarity-based multilingual symptom routing model
- A conversational chatbot prototype
- REST API for medical inference
- A web-based demonstration interface

12 Conclusion

This work presents a multilingual clinical department routing system that combines a structured symptom-matching model with a conversational chatbot interface.

By leveraging a curated dataset and similarity-based retrieval methods, the system enables preliminary healthcare guidance based on natural language symptom descriptions.

The hybrid architecture demonstrates a practical approach for integrating conversational AI with structured medical inference pipelines. Future work may explore transformer-based embeddings, multilingual training data, and voice-based symptom input.

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References